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Multi-Objective Optimization of the Multimodal Routing Problem Using the Adaptive ε-Constraint Method and Modified TOPSIS with the D-CRITIC Method

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Abstract: This paper proposes a multi-criteria decision-making approach for the multimodal routing problem (MRP) of bulk transportation in Thailand to minimize the total cost, transportation time, and total carbon dioxide-equivalent (CO_2e) emissions simultaneously. The proposed approach has three phases: The first phase is generating all nondominated solutions using Kirlik and Sayin's adaptive ε-constraint method. In the second phase, the Distance Correlation-based Criteria Importance Through Inter-criteria Correlation (D-CRITIC) method is used to determine the weight of each objective function and assign it to the modified technique for order of preference by similarity to ideal solution (modified TOPSIS) model in next phase. The third phase consists of ranking Pareto solutions obtained from the first phase using the modified TOPSIS. This proposed approach is applied to a real-world problem to enable the selection of the best route for transporting goods from the anchorage area in the Gulf of Thailand to the destination factory throughout a multimodal transportation network in Thailand. The computational results indicate that the proposed approach is superior to the current approach utilizing the ε-constraint method (ECM) regarding the number of Pareto solutions obtained and the proportion of computational time to the number of Pareto solutions obtained. Finally, the proposed method can solve the MRP with three or more objective functions and provide a multimodal route selection approach that is suitable for decision makers to offer a multimodal route to customers in the negotiation process for outsourcing transportation.

Keywords: multimodal transportation; adaptive ε-constraint method; multi-objective optimization; modified TOPSIS; multimodal routing problem; D-CRITIC

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1. Introduction

Multimodal transportation is a type of transportation involving at least two modes of transport from the origin to the destination under a single contract [1]. This type of transportation is key to ensuring efficient transportation in terms of environmental friendliness and the timely availability of products and raw materials [2]. For this reason, an efficient optimization approach to solve the multimodal routing problem (MRP) can improve overall supply chain efficiency.

The literature on MRP is extensive and growing [3]. Despite the abundant literature in this field, the emergence of multi-objective multimodal transport routing models and methods to address this issue have only been adopted recently [4]. Yang et al. [5] introduced a goal programming model aimed at minimizing transportation costs and travel time for the intermodal freight routing problem. Verma et al. [6] formulated a bi-objective optimization model that reduces the transport risk and cost for routing the rail–truck intermodal transportation of hazardous materials. They also developed a tabu search methodology to tackle instances of practical problems. Kengpol et al. [7] employed zero-one goal programming (ZOGP) that integrated the weights obtained from the Analytic Hierarchy Process (AHP) to solve the multimodal routing problem to minimize the cost, risk, and time. Xiong and Wang [8] presented a bi-level multi-objective Taguchi evolutionary algorithm to solve the

multimodal routing problem with time windows to reduce the total transportation cost and time. Kengpol et al. [9] established a framework using zero-one goal programming and AHP for the multimodal transportation route selection problem to minimize the cost, lead time, risk, and CO_2 emissions. Sun and Lang [10] formulated a bi-objective mixed integer linear programming model for a multi-modal routing problem to minimize the total transportation cost and the total transportation time. They applied the normalized normal constraint method to obtain the Pareto frontier of the bi-objective multi-modal routing problem. Assadipour et al. [11] developed a multi-objective genetic algorithm-based solution method for the rail–truck intermodal transportation problem of hazardous materials to minimize the total cost and public risks. Resat and Turkay [12] formulated a mixed-integer programming model for an intermodal transportation network design problem to minimize transportation costs and time simultaneously and to solve the problem using the augmented ε-constraint method. Baykasoğlu and Subulan [13] presented a mixed-integer mathematical programming model for a multi-objective intermodal freight transportation planning problem to minimize the total cost, transit time, and CO_2 emissions. Four multi-objective optimization methodologies have been employed to address conflicting objectives in both crisp and fuzzy decision-making contexts. Sun et al. [14] formulated a bi-objective mixed-integer linear programming model for the multimodal routing problem of hazardous material transportation to minimize the total costs and social risks. The normalized weighted sum method was used to generate Pareto solutions. Abbassi et al. [15] formulated a bi-objective mathematical model for an intermodal transportation problem of agricultural products to minimize transportation costs and maximize overtime to deliver products. Their model was solved with two solution approaches: a non-dominated sorting genetic algorithm (NSGA) improved by a local search heuristic and a greedy randomized adaptive search procedure (GRAS) with iterated local search heuristics (ILS). Sun et al. [16] proposed the combined fuzzy credibility chance constraint, linearization technique, and normalized weighting sum method for the road–rail multimodal routing problem under the uncertainty of hazardous materials to minimize the total cost and risks. Demir et al. [17] introduced a bi-objective linear mixed-integer mathematical formulation to address the green intermodal service network design problem to minimize both the total cost and CO_2e emissions. They applied the basic weighting sum method, the weighting sum method with normalization, and the ε-constraint method to generate Pareto solutions. Chen et al. [18] formulated a multi-objective optimization model to minimizing the total transportation cost, time, and container usage cost, using the normalized normal constraint method (NNCM) to obtain Pareto solutions. Sun et al. [19] applied the ε-constraint method to obtain the Pareto frontier of the fuzzy multi-objective routing problem with an uncertain demand and improved the service level in the road–rail multimodal network. Liaqait et al. [20] presented a fuzzy multi-criteria decision-making approach for the supplier selection and order allocation problem to minimize the environmental impact, total cost, equivalent sound level, total value of sustainable purchasing, and total travel time in the multi-modal transportation network. Koohathongsumrit and Meethom [21] proposed an integrated fuzzy risk assessment model (FRAM) and data envelopment analysis (DEA), including fuzzy AHP and ZOGP for the route selection problem in multi-modal transportation networks, to minimize the transportation cost, travel time, and total risk magnitude. Zhu and Zhu [22] proposed the K-shortest paths algorithm and the non-dominated sorting algorithm (NSGA-II) for the multi-objective route decision problem of multimodal transport under transport speed uncertainty and different carbon policies. Koohathongsumrit and Meethom [23] proposed an integrated AHP, DEA, and technique for order of preference by similarity to ideal solution (TOPSIS) for the route selection problem in multimodal transportation networks to minimize transportation cost, time, various risks, and environmental impact. Zhang et al. [24] proposed the multi-objective ant colony algorithm to solve a multimodal transport route selection model to maximize transport efficiency and to minimize cost, carbon emission, damage compensation costs, transportation reliability, and the time penalty cost. Koohathongsumrit and Chankham [25] proposed an integrated approach of the fuzzy

risk assessment-based incenter of centroid method, fuzzy AHP, and Vlse Kriterijumska Optimizacija I Kompromisno Resenje (VIKOR) for multimodal transportation route selection problems to minimize the travel cost, travel time, and various risks. Shao et al. [26] proposed a decision process based on NSGA-III and a dominance-based rough set approach (DRSA) to solve the customer-centered intermodal freight routing problem to minimize the total cost and maximize the objectives of punctuality, reliability, and flexibility.

Table 1 shows that there has been an increase in research on multi-objective methodologies for multimodal routing problems. The most frequently considered objective functions are cost, transit time, risk, carbon emissions, and other objective functions, respectively. The various and conflicting objectives make the problem more realistic for multimodal transportation planning applications [27]. There is a tendency for the MRP to have an increasing number of objectives, making this problem more challenging to compute [28]. In addition, the author found that the integration of MADM and MODM used to solve MRP with three or more objective functions is becoming more popular. Therefore, decision makers need effective multi-criteria decision making (MCDM) to solve real-world problems dealing with three or more objectives for the MRP. It is challenging to find the best solution for relevant methodologies, and more research is needed [3,27,29].

Table 1. An overview of articles dealing with multimodal routing problems.

Ref.	Objective Function					Solution Methodology	
	Cost	Time	COE	Risk	Other	MADM	MODM
Yang et al. [5]	√	√	-	-	-	-	GP
Verma et al. [6]	√	-	-	√	-	-	TSM
Kengpol et al. [7]	√	√	-	√	-	AHP	ZOGP
Xiong and Wang [8]	√	√	-	-	-	-	MOTGA
Kengpol et al. [9]	√	√	√	√	-	AHP	ZOGP
Sun and Lang [10]	√	√	-	-	-	-	WMN
Assadipour et al. [11]	√	-	-	√	-	-	MOGA
Resat and Turkay [12]	√	√	-	-	-	-	AECM
Baykasoğlu and Subulan [13]	√	√	√	-	-	-	CP, FGP-DIP, WFGP, IFGP
Sun et al. [14]	√	-	-	√	-	-	WMN
Abbassi et al. [15]	√	√	-	-	-	-	NSGA improved by GRAS, ILS
Sun et al. [16]	√	-	-	√	-	-	FCC, LT, WMN
Demir et al. [17]	√	-	√	-	-	-	WM, WMN, ECM
Chen et al. [18]	√	√	-	-	-	-	NNCM
Sun et al. [19]	√	-	-	√	-	-	ECM
Liaqait et al. [20]	√	√	√	-	√	Fuzzy AHP, Fuzzy TOPSIS, CRITIC, TOPSIS	AECM 2
Koohathongsumrit and Meethom [21]	√	√	-	√	-	Fuzzy AHP, DEA	ZOGP
Zhu and Zhu [22]	√	√	-	-	-	-	NSGA-II
Koohathongsumrit and Meethom [23]	√	√	√	√	-	AHP, DEA, TOPSIS	-
Zhang et al. [24]	√	√	√	√	√	-	MACA
Koohathongsumrit and Chankham [25]	√	√	-	√	-	Fuzzy AHP, VIKOR	-
Shao et al. [26]	√	√	-	-	√	DRSA	NSGA-III
This paper	√	√	√	-	-	D-CRITIC, Modified TOPSIS	Adaptive ECM

Note: COE = CO_2 emissions, MADM = multiple-attribute decision-making method, MODM = multiple-objective decision-making method, WSM = weighting sum method, WMN = weighting sum method with normalization, ECM = ε-constraint method, NSGA = non-dominated sorting genetic algorithm, ILS = iterated local search, GRAS = greedy randomized adaptive search procedure, CP = compromise programming technique with different distance metrics, FGP-DIP = fuzzy goal programming approach with different importance and priorities, WFGP = weighted additive fuzzy goal programming approach, IFGP = interactive fuzzy goal programming approach, AECM = augmented ε-constraint method, MOGA = multi-objective genetic algorithm, AHP = analytic hierarchy process, GP = goal programing model, ZOGP = zero-one goal programming, TSM = tabu search methodology, GA = genetic algorithm, NNCM = normalized normal constraint method, CRITIC = CRITIC weight method, SWARA = step-wise weight assessment ratio analysis method, DEA = data envelopment analysis, VIKOR = Vlse Kriterijumska Optimizacija I Kompromisno Resenje, DRSA = dominance-based rough set approach, MOTGA = multi-objective Taguchi genetic algorithm, TOPSIS = technique for order of preference by similarity to ideal solution modified method, FCC = fuzzy credibility chance constraint, LT = linearization technique, MACA = multi-objective ant colony algorithm.

The MCDM for addressing multi-objective transportation sector problems can be categorized into two categories: the multiple-attribute decision-making method (MADM) and the multiple-objective decision-making method (MODM) [30,31]. MADM is used to solve problems requiring selection from discrete sets of alternatives with known alternative solutions at the beginning. MODM is used for solving problems requiring selection from continuous sets of alternatives when alternative solutions are unknown at the beginning [31]. In recent years, several studies have proposed MADM and MODM to solve other problems in the transportation sector in addition to MRP. Tirkolaee et al. [32] proposed a fuzzy chance-constrained programming approach to solve the multi-trip location-routing problem with time windows in the COVID-19 pandemic. Lo et al. [33] proposed a performance calculation technique of the integrated multiple multi-attribute decision making (PCIM-MADM) and the modified indifference threshold-based attribute ratio analysis (ITARA) to solve transportation planning and the supplier selection problem. Aktar et al. [34] developed a type-2 fuzzy random goal programming method for a multi-objective green four-dimensional transportation problem. Chhibber et al. [35] proposed an intuitionistic fuzzy TOPSIS to solve a multi-objective nonlinear fuzzy transportation and manufacturing problem. Goli et al. [36] applied a goal programming method and robust optimization approach for an IoT-based sustainable supply chain network design with flexibility. Babaei et al. [37] developed entropy-TOPSIS, Nash equilibrium points, and data envelopment analysis to solve the transportation route selection problem with traffic flow. Tirkolaee et al. [38] developed Grey Wolf Optimization and Particle Swarm Optimization to solve a two-echelon multi-product location allocation routing problem. Simić et al. [39] proposed a multi-criteria decision-making method using a type-2 neutrosophic number based on the indifference threshold-based attribute ratio analysis method and the evaluation and distance from the average solution method to solve the route selection problem. Tirkolaee et al. [40] proposed an integrated approach with the multi-objective simulated annealing algorithm and the multi-objective invasive weed optimization algorithm. They utilized the Taguchi technique to optimally adjust the parameters and address the periodic capacitated arc routing problem. Wang et al. [41] applied the ε-constraint method for multi-objective transportation route optimization in hazardous material transportation. Oudani et al. [42] applied the weighting sum methods and the ε-constraint method to generate Pareto solutions and used several MADM approaches to rank solutions for blockchain technology for synchromodal transportation. According to literature reviews, most research focuses on the various proposed MCDMs to solve fuzzy problems in the transportation sector. However, classical methods such as the ε-constraint method are still effectively applied to generate Pareto solutions for transport problems.

The author found that many studies used classical weighting methods to determine the weights for each criterion in MADM to rank alternative routes and select the optimal route. Weighting methods are divided into three categories: subjective, objective, and integrated [43]. Subjective methods require preliminary information from decision makers before the weights are assigned to the criteria [44]. However, such information may be inappropriate for decision making in some cases. Decision makers' past beliefs lead to biased outcomes [45], and decision makers may lack in-depth knowledge of the problem under consideration, so there is an inability to provide the necessary preliminary information [46]. In contrast, objective methods do not require an initial input from decision makers, but instead employ a mathematical algorithm to determine the weights of the criteria [43,47]. This category of methods can eliminate possible biases associated with subjective assessments [48]. An integrated method is a combination of both subjective and objective methods [49,50]. In recent years, many new weighing methods have been proposed as follows [43]: the best–worst method (BWM) [51], Criterion Impact Loss (CILOS) [52], Integrated Determination of Objective CRIteria Weights (IDOCRIW) [52], the Full Consistency Method (FUCOM) [53], Level Based Weight Assessment (LBWA) [54], Simple Aggregation of Preferences Expressed by Ordinal Vectors Group Decision Making (SAPEVO-M) [55], MEthod based on the Removal Effects of Criteria (MEREC) [56], and D-CRITIC [44].

Currently, multimodal transport is a widely used platform for bulk cargo transportation in Thailand. The multimodal transport operator (MTO) is a decision maker responsible for solving the MRP to reduce costs, transportation time, and CO_2e emissions. Typically, the MTO must prepare a list of alternative routes for customers before negotiations, with preliminary information from the customer, including the origin, destination, and type of goods, before negotiating with the customer to agree on the price, delivery times, CO_2e emissions, and transportation route. When negotiating with customers, customer conditions may change at any time, and the MTO must immediately decide which route best suits the customer's conditions. A contract will be entered into if both parties agree to the transport job; otherwise, the MTO will not be employed for transportation. The problem that the MTO faces is seen in the current approach to solving the MRP, which provides too few alternative routes for customers. There is a high chance that no alternative route may satisfy the customer's conditions, making negotiating more challenging. The above reasons indicate that the current solution to the case study needs to be improved to be more effective, in line with research gaps in the literature review.

This paper proposes an approach based on the adaptive ε-constraint method of Kirlik and Sayın [57] and the modified TOPSIS with the D-CRITIC method for the multimodal routing problem (MRP), aiming to minimize the total cost, transportation time, and total carbon dioxide-equivalent (CO_2e) emissions. The proposed approach was tested with a real problem for bulk cargo transportation in Thailand, and the effectiveness of the proposed approach was examined by comparing it with the current approach based on the ECM. Additionally, the proposed approach provides a list of alternative routes to support selecting the best route for negotiating with customers, offering superiority over existing approaches to the MRP as follows: (1) The adaptive ε-constraint method in the proposed approach can guarantee that all Pareto solutions will be received and provides decision makers with more choices in presenting customers during negotiations. (2) The adaptive ε-constraint method is based on the current method (ε-constraint method) used in the case study to solve the problem, thus making it easy to implement the proposed approach for real-world applications. (3) The D-CRITIC method within the proposed approach was developed to detect linear and nonlinear relationships between two criteria, whereas the classical CRITIC method only detects linear relationships. The remaining content of the paper is structured as follows. Section 2 describes the multimodal transport routing problem. Section 3 proposes a multi-objective decision-making approach. Section 4 reports the computational results. The conclusion and directions of future research are outlined in Section 5.

2. Problem Description

The multimodal routing problem (MRP) aims to find routes for transporting raw materials or goods from the origin to the destination in the multimodal transport network under a single contract [10]. Three objective functions of this problem include minimizing the total cost, transportation time, and CO_2e emissions. Each terminal's transport mode and infrastructure must be consistent [58]. In this paper, let G (V, E, M) denote the multimodal transportation network, where V, E, and M are a set of nodes (terminals), transportation links, and transportation modes, respectively. The assumptions of the problem are as follows:

- Each transportation link can use only one mode of transport;
- The MTO cannot choose the transport routes until negotiations have been conducted with the customer, because the chosen route has to satisfy unpredictable customer conditions;
- The alternate routes are not explicitly defined;
- Only one job is carried out throughout the multimodal transportation network;
- The number of times a shipment of products must cross a node does not exceed one;
- All nodes, links, and terminals have sufficient capacity to handle a given transport or cargo operation.

The sets, parameters, and decision variables in the integer linear programming model are defined as follows.

Sets:	
V	a set of nodes (terminals) $i \in V, j \in V$;
E	a set of transportation links $(i, j) \in E$;
M	a set of transportation modes $m \in M$.
Parameters:	
c_{ij}^m	Transportation cost from terminals i to j via transportation mode m (Baht/ton);
h_i	Transshipment cost at terminal i (Baht/ton);
t_{ij}^m	Transportation time from terminals i to j via transportation mode m (hours);
s_i	Service time at terminal i (hours);
o_{ij}^m	CO_2e emissions from terminals i to j via transportation mode m (kgCO_2e);
e_i	CO_2e emissions at terminal i (kgCO_2e);
0	Origin terminal;
d	Destination terminal;
σ	Large number (σ = 999,999).
Decision Variables:	
x_{ij}^m	Binary decision variables: 1 = if the goods are transported from terminals i to j via transportation mode m and transshipped at terminal i; 0 = otherwise.

The multi-objective integer linear programming model for the MRP can be formulated as follows:

$$Min\ f_1(x) = Min \sum_{(i,j)\in E} \sum_{m\in M} \left[\left(h_i + c_{ij}^m \right) x_{ij}^m \right] \tag{1}$$

$$Min\ f_2(x) = Min \sum_{(i,j)\in E} \sum_{m\in M} \left[\left(s_i + t_{ij}^m \right) x_{ij}^m \right] \tag{2}$$

$$Min\ f_3(x) = Min \sum_{(i,j)\in E} \sum_{m\in M} \left[\left(e_i + o_{ij}^m \right) x_{ij}^m \right] \tag{3}$$

where

$$c_{ij}^m = \begin{cases} \sigma, & i = j \\ c\prime_{ij}^m, & otherwise \end{cases} \tag{4}$$

$$t_{ij}^m = \begin{cases} \sigma, & i = j \\ t\prime_{ij}^m, & otherwise \end{cases} \tag{5}$$

$$o_{ij}^m = \begin{cases} \sigma, & i = j \\ o\prime_{ij}^m, & otherwise \end{cases} \tag{6}$$

subject to

$$\sum_{j\in V} \sum_{m\in M} x_{ij}^m - \sum_{i\in V} \sum_{m\in M} x_{ji}^m = \begin{cases} 1, i = 0 \\ -1, i = d \\ 0, otherwise \end{cases} \quad \forall i \in V \tag{7}$$

$$\sum_{m\in M} x_{ij}^m \leq 1 \quad \forall (i,j) \in E \tag{8}$$

$$\sum_{i\in V} \sum_{j\in V} x_{ij}^m \leq 1 \quad \forall m \in M \tag{9}$$

$$x_{ij}^m \in \{0,1\} \quad \forall (i,j) \in E, m \in M \tag{10}$$

In this model formulation, Equation (1) is the first objective function, for minimizing the total cost. The second objective function minimizes the total transportation time in Equation (2), and the last objective function minimizes the total CO_2e emissions in Equation (3). Equation (4) describes the transportation cost from terminals i to j via transportation mode m. Equations (5) and (6) describe the transportation time from terminals I to j via transportation mode m and CO_2e emissions from terminals i to j via transportation

mode m, respectively. CO_2e emissions e_i and o_{ij}^m were computed from the emission factors of various transportation modes using the emission factor estimation method following the Thailand Greenhouse Gas Management Organization (TGO) [59,60]. Constraint (7) is the flow equilibrium constraint for each terminal. Constraint (8) ensures that one transportation mode m, at most, is used on the transportation link (i, j). Constraint (9) ensures that one transportation mode m for one route is selected. Constraint (10) represents the integrality conditions of the decision variables.

3. Proposed Approach

The proposed approach aims to provide a list of alternative routes with the ranking of each alternative route to support the selection of the best route. The author attempts to integrate MADM, MODM, and the weight determination method to solve the problem. The method chosen in the proposed approach has been proven to be more effective than the classical method.

The proposed approach comprises three phases. The first phase involves generating all Pareto solutions using Kirlik and Sayin's adaptive ε-constraint method [57], as described in Section 3.1. In the second phase, the D-CRITIC method [44] is used to determine the weight of each objective function, as described in Section 3.2. In the third phase, the modified TOPSIS method [61] ranks Pareto solutions or alternative routes, as demonstrated in Section 3.3. A flow chart of the proposed approach is illustrated in Figure 1.

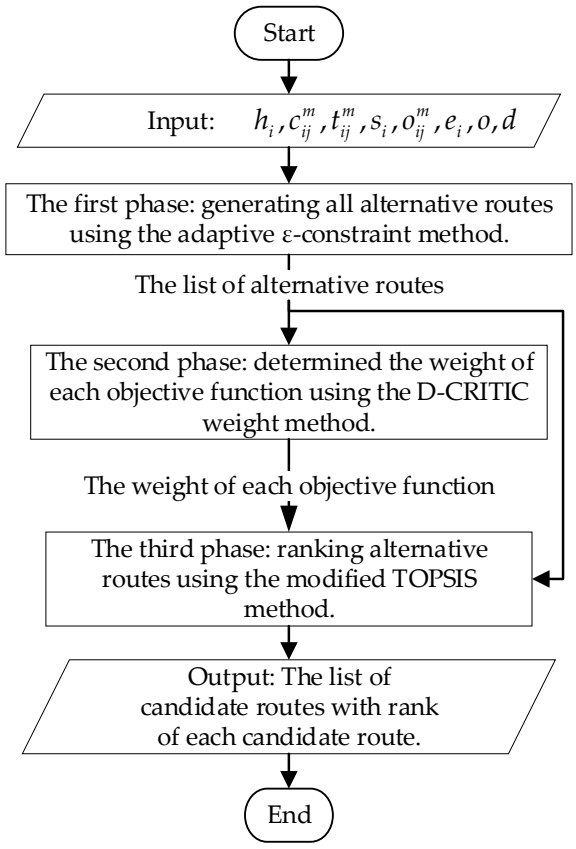


Figure 1. A flow chart of the proposed approach.

3.1. Generating All Pareto Solutions

This phase involves generating all alternative routes or Pareto solutions using Kirlik and Sayin's adaptive ε-constraint method [57]. The adaptive ε-constraint method of Kirlik and Sayın is the ECM with a search mechanism based on the generation of rectangles, as in Laumanns et al. [62], and some of the rectangles can be removed if no Pareto solution can be found. The set of feasible outcomes in the objective space is defined as each feasible

solution $x \in X$, which is mapped into its corresponding objective vector $y = f(x)$. Since ε is in $(P-1)$ dimensional space, the search is managed in $\mathbb{R}^{P-1}$.

This method is classified as an exact solution to multi-objective integer optimization problems with more than three objective functions [63]. This allows for decision makers to obtain the most significant number of alternative routes. This method was selected for the following reasons: (1) This method allows for decision makers to obtain all Pareto solutions or alternative routes. As a result, decision makers have access to as many optimal routes as possible for customers to consider during negotiations of outsourcing transportation operations. (2) Currently, MTOs already use ECM. If they switch to using this method based on the ECM, it will make it easy for planners to understand the principles of solving the problem, because this method is based on the ECM as well. (3) From the computational results in the article by Kirlik and Sayın [57], it can be seen that this method is more efficient than previous algorithms in terms of both computational time and the number of models solved per Pareto solution. A flow chart of Kirlik and Sayin's adaptive ε-constraint method is illustrated in Figure 2.

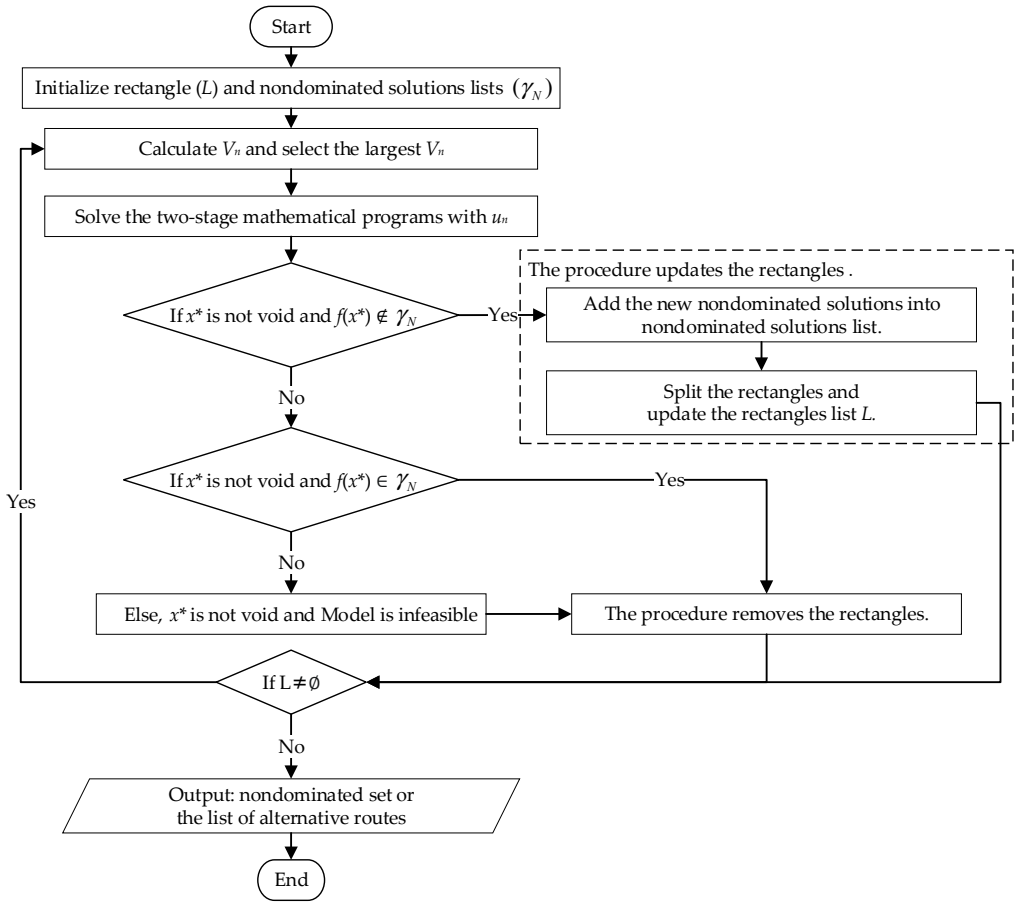


Figure 2. A flow chart of Kirlik and Sayin's adaptive ε-constraint method [57].

The solution steps of Kirlik and Sayin's adaptive ε-constraint method [57] are as follows.

Step 1: Find the lower bound of a rectangle (y_p^l) by minimizing each single-objective function for $p = \{2, \ldots, P\}$ as in Equation (11), subject to constraints (7)–(10).

$$y_p^l = Min\ f_p(x) \tag{11}$$

subject to Constraints (7)–(10).

Step 2: Find the upper bound of a rectangle (y_p^u) by maximizing each single-objective function for all $p \in \{2, \ldots, P\}$ as in Equation (12), subject to constraints (7)–(10).

$$y_p^u = Max\ f_p(x) \tag{12}$$

subject to Constraints (7)–(10).

The definitions of t_{ij}^m and o_{ij}^m in Equations (5) and (6) can be changed to that in Equations (13) and (14) to ensure that the solution of the upper bound is possible.

$$t_{ij}^m = \begin{cases} 0, & i = j \\ tr_{ij}^m, otherwise \end{cases} \tag{13}$$

$$o_{ij}^m = \begin{cases} 0, & i = j \\ or_{ij}^m, otherwise \end{cases} \tag{14}$$

The lower bound $\left(y_2^l, y_3^l\right)$ and upper bound $\left(y_2^u, y_3^u\right)$ are provided from steps 1 and 2. Then, $\overline{y}^l$ and $\overline{y}^u$ are defined in $\mathbb{R}^{P-1}$, where $\overline{y}^l = \left(y_2^l, \ldots, y_p^l\right)$ and $\overline{y}^u = \left(y_2^u, \ldots, y_p^u\right)$. Therefore, an initial rectangle can be defined as $L \leftarrow \left\{R\left(\overline{y}^l, \overline{y}^u\right)\right\}$. The lower and upper bounds of the $(P-1)$-dimensional rectangle are shown in Figure 3.

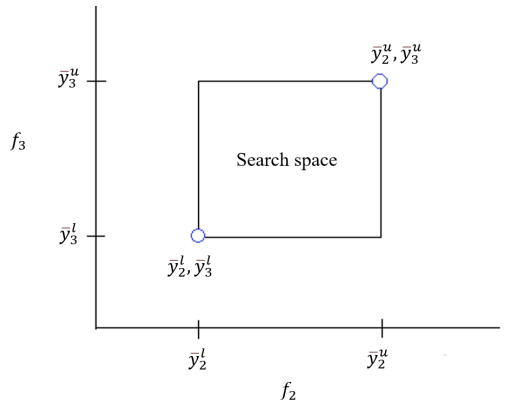


Figure 3. The projected points of the lower and upper forms of the $(P-1)$ dimensional rectangles in the search space.

Step 3: Calculate the volume measure associated with all rectangles for iteration n (V_n), as shown in Equation (15). Then, a rectangle with a larger V_n is selected.

$$V_n = \prod_{p=1}^{P-1}\left(u_{np} - \overline{y}_p^l\right) \tag{15}$$

Step 4: The objective function p is kept as the objective function in $Kq(\varepsilon)$ (in this paper, the objective function is set as q = 1), while the other $P-1$ values are converted into constraints [64]. The ε_p values are on the $f_2(x) - f_3(x)$ plane and the projection of all Pareto solutions in $\mathbb{R}^{3-1}$. $l \in \mathbb{R}^{P-1}$ and $u \in \mathbb{R}^{P-1}$ are defined as the lower and upper vertices of the rectangle, respectively. Therefore, a rectangle is defined as $R(l,u) = \left\{\overline{y} \in \mathbb{R}^{P-1}, l \le \overline{y} \le u\right\}$ [65]. The two-stage mathematical programs $K_1(u_n)$ and $Q_1(u_n)$ are solved using the selected rectangle's upper vertex u_n as ε. For each rectangle, two-stage optimization problems are solved to avoid weakly efficient solutions, and $K_q(\varepsilon)$ and $Q_q(\varepsilon)$ are defined as follows:

$$K_q(\varepsilon)\ \ \min f_q(x) \tag{16}$$

subject to

$$f_p(x) \leq \varepsilon_p \quad \forall p \in P, q \neq p \tag{17}$$

Constrains (7)–(10).

Consider z^* of the sub-problem $K_q(\varepsilon)$ in the second stage formulation $Q_q(\varepsilon)$. Define x^* as the optimal solution of two-stage formulations, with x^* always efficient for any $\varepsilon \in \mathbb{R}^{P-1}$.

$$Q_q(\varepsilon) \quad \min \sum_{p=1}^{P} f_p(x) \tag{18}$$

subject to

$$f_p(x) \leq \varepsilon_p \qquad \forall p \in P, q \neq p \tag{19}$$

$$f_q(x) = z^* \tag{20}$$

Constrains (7)–(10).

Step 5: If an optimal solution of two-stage formulations (x^*) is not void, and $f(x^*)$ is not a member in the Pareto solutions list (γ_N), x^* is a new Pareto solution and can be added into γ_N ($\gamma_N \leftarrow \gamma_N \cup \{f(x^*)\}$). Otherwise, step 8 is followed.

Step 6: $\overline{f}_p(x^*)$ is defined as the projection of the new Pareto solution for objective function p. Split the rectangles that satisfy $l_n < \overline{f}_p(x^*) < u_n$, and then rectangle R_n can be split into two rectangles along the pth axis. The first resulting rectangle has $\overline{y}_p^l \leq \overline{f}_p(x^*)$. The second resulting rectangle has $\overline{y}_p^l \geq \overline{f}_p(x^*)$. Then, the list of rectangles L can be updated. The pseudo-code for the procedure updates the rectangles, as shown in Figure 4.

The procedure updates the rectangles.

Input: List of rectangle L, projection of the new Pareto solution $\bar{f}_p(x^*)$.
Output: Updates list of rectangle L
$L' \leftarrow L, L \leftarrow \emptyset$
Foreach $(R_n \in L')$ do
  $T \leftarrow \{R_n(l_n, u_n)\}$
  For $p = 1$ to $P - 1$ do
    If $(l_{np} < \bar{f}_p(x^*) < u_{np})$ then
      /* Split all rectangles in T.
      $T' \leftarrow \emptyset$
      Foreach $(R_t \in T)$ do
        /* Split the rectangle R_t into two new rectangles ($\hat{R}$ and R').
        /* First resulting rectangle.
        $\hat{R} = \{\bar{y} \in R_t : \bar{y}_j \leq \bar{f}_j(x^*)\}$
        /* Second resulting rectangle.
        $R' = \{\bar{y} \in R_t : \bar{y}_j \geq \bar{f}_j(x^*)\}$
        $T' \leftarrow T' \cup \{\hat{R}\} \cup \{R'\}$
      End
      $T \leftarrow T'$
    End
  End
  $L \leftarrow L \cup T$
End
Return L

Figure 4. The procedure updates the rectangles [57].

Step 7: The rectangles in which the volume between the upper vertex of the rectangle and $\overline{f}(x^*)$ do not need to be searched when x^* is an optimal solution of two-stage programs

$P_1(u_n)$ and $Q_1(u_n)$ are removed. Then, there no Pareto solution is mapped into $R\left(\overline{f}\left(x^*\right), u_n\right)$ other than $f(x^*)$. The pseudo-code for the procedure removes the rectangles, as shown in Figure 5.

The procedure removes the rectangles.
Input: List of rectangle L, lower vertex l, upper vertex u. Output: Updates list of rectangle L Define a rectangle $R(l,u)$ Foreach $(R_t \in L)$ do   If $(R_t \subseteq R(l,u)$ then $L \leftarrow L/\{R_t\}$ End End Return L

Figure 5. The procedure removes the rectangles [57].

Step 8: If x^* is not void, and $f(x^*)$ is a member in the Pareto solution list (γ_N), remove the rectangle $R\left(\overline{f}\left(x^*\right), u_n\right)$ for iteration n using the procedure, which removes the rectangles in step 7. Otherwise, the next step is employed.

Step 9: If x^* is void, or the model is infeasible, there is no Pareto solution that is mapped into the rectangle $R\left(\overline{y}^l, u_n\right)$; all such rectangles will be removed using the procedure that removes the rectangles in step 7.

Step 10: If the list of rectangles is empty ($L \neq \emptyset$), the algorithm is ended and returns all Pareto solutions or all alternative routes. Otherwise, step 3 is followed.

3.2. Determination of the Weight of Each Objective Function

The weight of each objective function can be determined using the D-CRITIC method. The D-CRITIC method [44] was developed by incorporating the concept of distance correlation into the traditional CRITIC method [66] to improve the shortcomings in denoting the conflicting relationships between the criteria of the Pearson's correlation [67] in the original CRITIC method [44,68]. The steps of the D-CRITIC method are as follows.

Step 1: Obtain all alternative routes from the first phase to generate the decision matrix, as shown in Table 2. r is defined as the alternative route ($r \in AR$), c is the criterion or objective function ($c \in C$, in this paper $C = 3$), and a_{rc} is the objective function $f(x)$ of alternative route r with respect to criterion c.

Table 2. The decision matrix.

Alternatives/Criteria	Criteria_1 [1]	Criteria_2 [2]	Criteria_3 [3]
Alternative route no. 1	a_{11}	a_{12}	a_{13}
Alternative route no. 2	a_{21}	a_{22}	a_{23}
⋮	⋮	⋮	⋮
Alternative route AR	a_{R1}	a_{R2}	a_{R3}

[1] Total cost, [2] total time, [3] total CO_2e emissions.

Step 2: Normalize the decision matrix, using Equation (21) to transform a_{rc} into standard scales, which range between 0 and 1:

$$\overline{a_{rc}} = \frac{a_{rc} - a_c^{worst}}{a_c^{best} - a_c^{worst}} \quad (21)$$

where $\overline{a_{rc}}$ is the normalized score of alternative route r with respect to criterion c, a_c^{best} is the best objective function of criterion c, and a_c^{worst} is the worst objective function of criterion c.

Step 3: Calculate the standard deviation of criterion c using Equation (22). Define $\overline{a_c}$ as the mean objective function of criterion c, and AR is the total number of alternatives.

$$SD_c = \sqrt{\frac{\left(\sum_{r=1}^{R} a_{rc} - \overline{a_c}\right)^2}{AR - 1}} \tag{22}$$

Step 4: Calculate the distance correlation introduced by Székely et al. [68] for capturing the conflicting relationships between criteria c and criteria $c\prime$ using Equation (23) [44]:

$$dCor(\text{criteria}_c, \text{criteria}_{c\prime}) = \frac{dCov(\text{criteria}_c, \text{criteria}_{c\prime})}{sqrt(dVor(\text{criteria}_c)dVor(\text{criteria}_{c\prime}))} \tag{23}$$

where $dCov(\text{criteria}_c, \text{criteria}_{c\prime})$ is the distance covariance between criteria c and criteria $c\prime$, $dVor(\text{criteria}_c) = dCov(\text{criteria}_c, \text{criteria}_c)$ is the distance variance of criteria c, and $dVor(\text{criteria}_{c\prime}) = dCov(\text{criteria}_{c\prime}, \text{criteria}_{c\prime})$ is the distance variance of criteria $c\prime$ [69].

Step 5: Calculate the information content contained in criterion c (I_c) using Equation (24):

$$I_c = SD_j \sum_{c\prime=1}^{C} (1 - dCor(\text{criteria}_c, \text{criteria}_{c\prime})) \tag{24}$$

Step 6: Determined the final objective weight of criterion c using Equation (25). Define W_c as the objective weight of criterion c.

$$W_c = \frac{I_c}{\sum_{c=1}^{C} I_c} \tag{25}$$

3.3. Ranking Alternative Routes

The final phase of the proposed approach is to rank alternative routes using the modified TOPSIS method [61]. The main justifications for employing the modified TOPSIS method are as follows: (1) The modified TOPSIS method is relatively straightforward to understand and implement. Its simplicity allows for decision makers to easily interpret the results and make informed choices. (2) The method allows for an ability to provide a clear actionable ranking of alternatives based on multiple criteria. (3) The modified TOPSIS method considers both positive (best) and negative (worst) ideal solutions. This makes it more robust and applicable to real-world scenarios. The modified TOPSIS method [61] comprises the following steps.

Step 1: Normalize the decision matrix using Equation (26). Define b_{rc} as the normalized performance ratings of alternative route r with respect to criterion c.

$$b_{rc} = \frac{a_{rc}}{\sqrt{\sum_{r=1}^{AR} a_{rc}^2}} \tag{26}$$

Step 2: Calculate the positive ideal solutions (β^*) and negative ideal solutions (β^-) from the normalized decision matrix.

$$\beta^* = [b_1^*, b_2^*, \ldots, b_c^*] \tag{27}$$

$$\beta^- = \left[b_1^-, b_2^-, \ldots, b_c^-\right] \tag{28}$$

where

$$b_c^* = \min_{r \in R} b_{rc} \quad \forall c \in C \tag{29}$$

$$b_c^- = \max_{r \in R} b_{rc} \quad \forall c \in C \tag{30}$$

Step 3: Calculate the weighted Euclidean distance using Equations (31) and (32). Define WE_r^* as the positive weighted Euclidean distance, WE_r^- as the negative weighted Euclidean distance, and W_c as the objective weight of criterion c (obtained from Section 3.2).

$$WE_r^* = \sqrt{\sum_{c=1}^{C} W_c (b_{rc} - b_c^*)^2} \tag{31}$$

$$WE_r^- = \sqrt{\sum_{c=1}^{C} W_c (b_{rc} - b_c^-)^2} \tag{32}$$

Step 4: Calculate the overall performance score for each alternative route OP_r, as in Equation (33):

$$OP_r = \frac{WE_r^-}{WE_r^- + WE_r^*} \tag{33}$$

Step 5: Rank the competing alternative routes by utilizing the performance score. An alternative route r with higher scores indicates a better alternative performance [61,70]. Therefore, decision makers can present the alternative route with the highest performance score to the customer for consideration first.

4. Results

In this study, the proposed approach was applied to a real case study to select the best route for transporting bulk cargo (mainly coal) from the anchorage area in the Gulf of Thailand to the destination factory via a multimodal transportation network in Thailand. The multimodal transportation network has 83 nodes and 9579 transportation links, consisting of the load discharge area in the Gulf of Thailand (node 83), the destination factory (node 73), 12 nodes for the seaport, 32 nodes for the river port, 30 nodes for the warehouse, and 4 nodes for the railway station, as shown in Figure 6. There are four modes of transportation: dump trucks, barges, trains, and vessels. Each transport task has a capacity of 20,000 tons, equivalent to a vessel's capacity. Different types of lines were used to represent the mode of transportation. One mode of transportation could be used for one transportation link. Each transportation link has a different distance. The branch-and-cut algorithm in OpenSolver 2.9.0 was used with the default configuration settings to solve this problem on a computer with an AMD Ryzen 5 4600 H 3.00 GHz and 16 GB RAM. The collected instances and parameters from July to October 2022 are available for download through the following URL: https://sites.google.com/site/transportationlibrary/instance/p11 (accessed on 5 July 2023).

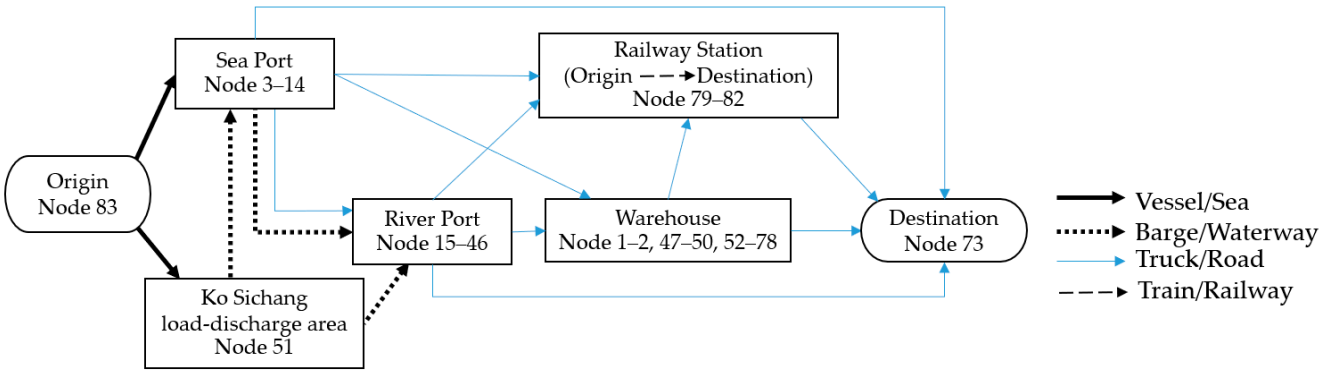


Figure 6. The multimodal transportation network in Thailand used in this case study.

Multimodal transport operators (MTOs) have to solve multi-objective multimodal routing problems with predetermined origins and destinations traveling through the multimodal transport network in Thailand using the ε-constraint method [64]. G_p was defined as the number of solutions for objective p to generate the Pareto solution (g_p = 1, 2, …, G_p). The decision makers can set the epsilon values ε_2 and ε_3 to extract the Pareto-optimal solution using Equation (34) [19].

$$\varepsilon_p = y_p^l + \frac{y_p^u - y_p^l}{G_p - 1} \cdot (g_p - 1) \quad \forall g_p \in G_p, \forall p \in P \tag{34}$$

In addition, there is no guideline for selecting the appropriate transportation route for customer negotiations. Thus, this current approach relies on decision maker experience in selecting routes from the Pareto solution set for customer negotiations. The MTO needs an approach that provides the most Pareto solutions and an appropriate route selection guideline for negotiating with the customer; there must be various Pareto solutions that decision makers can present to the customer.

4.1. Single-Objective Optimization Results

The first phase of the proposed method was initially solved by considering single objectives of $f_2(x)$ and $f_3(x)$ separately to determine the lower and upper bounds for assigning the epsilon values (ε_2, ε_3). The lower and upper bounds of each objective function were obtained from the exact linear solver, as shown in Table 3.

Table 3. The lower and upper bounds of each objective function.

Objective Function	Min $f_2(x)$	Min $f_3(x)$	Max $f_2(x)$	Max $f_3(x)$	Lower Bound	Upper Bound
$f_2(x)$	64	74	150	150	64	150
$f_3(x)$	273,945.41	215,171.92	751,698.36	751,698.36	215,171.92	751,698.36

Note: $f_2(x)$ = total time, $f_3(x)$ = total CO_2e emissions.

4.2. Multi-Objective Optimization Results

In this section, the Pareto fronts obtained from the proposed approach (PA) based on the adaptive ε-constraint method of Kirlik and Sayin [57] and the current approach based on the ε-constraint method (ECM) [64] with differences in the epsilon value settings were compared to assess the effectiveness of the proposed approach in terms of the number of Pareto solutions found (NPS) and the computational time (CPU). The three scenarios were defined in setting the epsilon values ε_2 and ε_3 in the ECM compared with the PA by changing the G_p values in Equation (34) as follows: situation 1 (ECM 4 × 4) is set as G_2 = 4 and G_3 = 4, situation 2 (ECM 6 × 6) is set as G_2 = 6 and G_3 = 6, and situation 3 (ECM 10 × 10) is set as G_2 = 10 and G_3 = 10. The Pareto solutions obtained and the computational time of the current and the proposed methods are presented in Table 4.

Table 4 demonstrates that the proposed approach can identify Pareto solutions more than ECM 4 × 4, ECM 6 × 6, and ECM 10 × 10 by 133.33%, 75.00%, and 16.67%, respectively. Moreover, the proposed method took the least computational time per Pareto solution compared to ECM 4 × 4, ECM 6 × 6, and ECM 10 × 10, by 15.52%, 51.96%, and 76.74%, respectively. The proposed method was more efficient than the current method for all experiments that changed the epsilon value settings in terms of the number of Pareto solutions found and the ratio of computational time to the number of Pareto solutions found. The Pareto frontier obtained from the proposed method indicates that increasing the total cost (f_1) decreases the total time (f_2) and the total CO_2e emissions (f_3), as shown in Figure 7.

Table 4. Comparison of the proposed and current approaches with differences in epsilon value settings.

PS no.	Route	Total Cost (Baht/Ton)	Total Time (Days)	Total CO_2e Emissions ($kgCO_2eq$)	ECM 4 × 4	ECM 6 × 6	ECM 10 × 10	PA
1	83 → 5 ⋯▸ 46 → 73	347.62	84	294,729.92	√	√	√	√
2	83 → 5 ⋯▸ 17 → 73	353.98	81	315,162.85		√	√	√
3	83 → 14 ⋯▸ 46 → 73	409.93	74	254,405.12			√	√
4	83 → 14 ⋯▸ 31 → 73	413.52	67	263,787.52			√	√
5	83 → 3 → 73	463.95	64	273,945.41	√	√	√	√
6	83 → 3 ⋯▸ 46 → 73	468.66	74	215,171.92	√	√	√	√
7	83 → 3 ⋯▸ 35 → 73	472.73	67	225,396.08				√
The number of Pareto solutions found					3	4	6	7
Computational time (second)					5.21	12.22	37.90	10.27
CPU/NPS					1.74	3.06	6.32	1.47

Note: √ = the Pareto solution can be found using that approach, PS no. = Pareto solution number, ECM 4 × 4 = G_2 is 4 and G_3 is 4 for the current approach, ECM 6 × 6 = G_2 is 6 and G_3 is 6 for the current approach, ECM 10 × 10 = G_2 is 10 and G_3 is 10 for the current approach, PA = proposed approach, CPU/TPS = the ratio of computational time to the number of Pareto solutions found. → = transport via vessel, ⋯▸ = transport via barge, → = transport via truck.

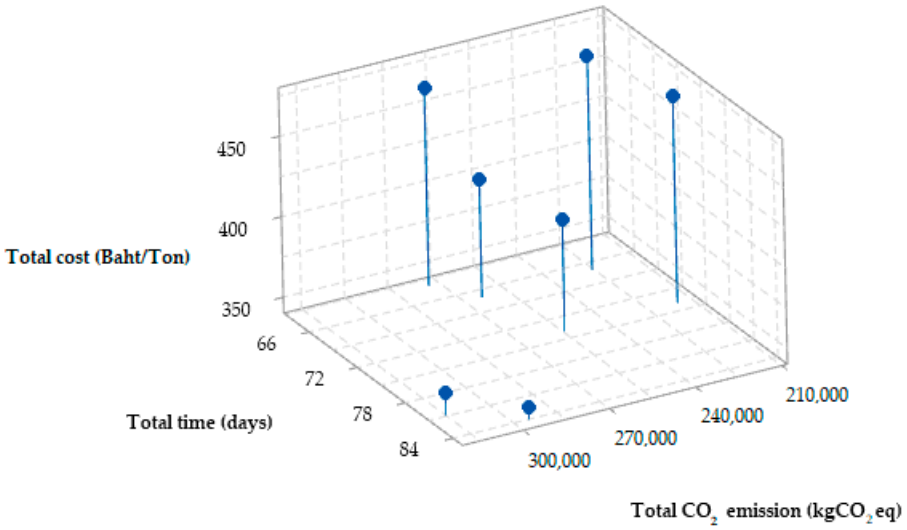


Figure 7. The Pareto frontier obtained from the proposed method for case study problem.

4.3. Pareto Solution Ranking Results

The results of the Pareto solution ranking were obtained using modified TOPSIS [61] with the D-CRITIC weighting method [44], identified in this section. The weights of three objective functions using the D-CRITIC method are presented in Table 5.

Table 5. Weight of three objective functions using the D-CRITIC method.

W_1	W_2	W_3
0.2970	0.3724	0.3307

Note: W_1 is the objective weight of the total cost, W_2 is the objective weight of the total time, and W_3 is the objective weight of the total CO_2e emissions.

After calculating the criteria weights, the next step was to apply modified TOPSIS [61] to calculate the overall performance score (OP_r) matrix. The results of the *OPr* matrix for Pareto solutions are presented in Table 6.

Table 6. The overall performance score (*OPr*) matrix for Pareto solutions of the proposed method.

PS no.	The Overall Performance Score	Rank
1	0.4117	6
2	0.3753	7
3	0.5501	4
4	0.5867	2
5	0.5671	3
6	0.4898	5
7	0.5929	1

Note: PS no. = Pareto solution number.

It is evident from Table 6 that Pareto solution number 7 had the highest overall performance score value, at 0.5929, among all seven solutions, having a total cost of 472.73 baht/tons, a total time of 67 days, and total CO_2e emissions of 225,396.08 $kgCO_2eq$. Therefore, Pareto solution no. 7 is the first choice that decision makers can present to their customers. If the customer is not interested in Pareto solution number 7, the decision maker will offer Pareto solution numbers 4, 5, 3, 6, 1, and 2, respectively.

5. Conclusions

This study applied Kirlik and Sayin's adaptive ε-constraint method [57] for generating all Pareto solutions, and then ranked the Pareto solutions using the modified TOPSIS [61] with the D-CRITIC weighting method [44] to solve the multimodal routing problem (MRP) of bulk transportation in Thailand, with the aim of minimizing the total cost, transportation time, and total carbon dioxide-equivalent (CO_2e) emissions simultaneously. For this case study, the decision maker had to choose an appropriate route from the Gulf of Thailand to the destination plant, through the multimodal transportation network in Thailand, for presentation to the customer during the negotiation process for outsourcing transportation.

The computational results indicate that the proposed approach exhibits superior performance to the current approach based on the ε-constraint method in terms of the number of Pareto solutions found and the ratio of computational time to the number of Pareto solutions found. In addition, the proposed approach also provides a rank for deciding the appropriate route for customers, whereas the current method applied to case studies still requires the decision maker's experience to choose a route to present to the customer.

From a theoretical perspective, this paper contributes the following: First, a novel multi-criteria decision-making approach was introduced for the MTRP with three or more objective functions. Second, this is the first time that the adaptive ε-constraint method of Kirlik and Sayın [57] has been applied to find all Pareto solutions for the MRP. This approach could be the basis for starting further work to solve the MRP.

Regarding practical implications, this paper proposes several implications on the practical side of things: First, the proposed approach provides assistance to decision makers in selecting the best route and ranks other alternative routes from different perspectives. Additionally, it increases the likelihood that the MTO will reach an agreement with its customers. Second, the integrated weighting method and MADM allow for practitioners and managers to prioritize alternatives quickly and efficiently without relying on expert experience or decision makers. MTOs can benefit significantly from this proposed approach, especially when decision makers require an accurate ranking of the possible alternative routes.

In the future, a novel approach should be developed for the fuzzy multimodal routing problem with uncertain transportation times due to traffic conditions and other factors to obtain the most suitable transportation routes and acceptable computational times. In addition, objective considerations of social impacts and risks need to be added to the mathematical model to enable more comprehensive decision making.

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Nomenclatures

Abbreviation	
AECM	Augmented ε-constraint method
AHP	Analytic hierarchy process
BWM	Best–worst method
CILOS	Criterion Impact Loss
CO_2e	Carbon dioxide-equivalent
COE	CO_2 emissions
CP	Compromise programming technique with different distance metrics
CPU	Computational time
CPU/NPS	Ratio of computational time to the number of Pareto solutions found
CRITIC	CRITIC weight method
D-CRITIC	Distance Correlation-based Criteria Importance Through Inter-criteria Correlation
DEA	Data envelopment analysis
DRSA	Dominance-based rough set approach
ECM	ε-constraint method
FCC	Fuzzy credibility chance constraint
FGP-DIP	Fuzzy goal programming approach with different importance and priorities
FUCOM	Full Consistency Method
GA	Genetic algorithm
GP	Goal programing model
GRAS	Greedy randomized adaptive search procedure
IDOCRIW	Integrated Determination of Objective CRIteria Weights
IFGP	Interactive fuzzy goal programming approach
ILS	Iterated local search
LBWA	Level Based Weight Assessment
LT	Linearization technique
MACA	Multi-objective ant colony algorithm
MADM	Multiple-attribute decision-making method
MEREC	MEthod based on the Removal Effects of Criteria
MODM	Multiple-objective decision-making method
MOGA	Multi-objective genetic algorithm
MOTGA	Multi-objective Taguchi genetic algorithm
MRP	Multimodal routing problem
MTO	Multimodal transport operator
NNCM	Normalized normal constraint method
NPS	The number of Pareto solutions found
NSGA	Non-dominated sorting genetic algorithm
PA	Proposed approach

SAPEVO-M	Simple Aggregation of Preferences Expressed by Ordinal Vectors Group Decision Making
SWARA	Step-wise weight assessment ratio analysis method
TGO	Thailand Greenhouse Gas Management Organization
TOPSIS	Technique for order of preference by similarity to ideal solution
TOPSIS	Technique for order of preference by similarity to ideal solution modified method
TSM	Tabu search methodology
VIKOR	Vlse Kriterijumska Optimizacija I Kompromisno Resenje
WFGP	Weighted additive fuzzy goal programming approach
WMN	Weighting sum method with normalization
WSM	Weighting sum method
ZOGP	Zero-one goal programming
Sets and Subscript	
c	The criterion or objective function ($c \in C$, in this paper $C = 3$)
E	The set of transportation links $(i, j) \in E$
G_p	The set of solutions for objective p to generate the Pareto solution ($g_p = 1, 2, \ldots, G_p$)
l	Lower bounds
M	The set of transportation modes $m \in M$
n	Iteration for search algorithm in adaptive ε-constraint method
P	The set of objective functions $p \in P$
q	The first objective functions ($q = 1, q \neq p$)
r	The alternative route ($r \in AR$)
u	Upper bounds
V	The set of nodes (terminals) $i \in V, j \in V$
X	The set of feasible solutions $x \in X$
Parameters and variables	
$\overline{a_c}$	The mean objective function of criterion c
a_c^{best}	The best objective function of criterion c
a_c^{worst}	The worst objective function of criterion c
a_{rc}	The objective function $f(x)$ of alternative route r with respect to criterion c
$\overline{a_{rc}}$	The normalized score of alternative route r with respect to criterion c
b_c^*	Alternative route with the highest normalized performance score in terms of criterion c
b_c^-	Alternative route with the minimum normalized performance score in terms of criterion c
b_{rc}	The normalized performance ratings of alternative route r with respect to criterion c
c_{ij}^m	Transportation cost from terminals i to j by transportation mode m (Baht/ton)
d	Destination terminal
$dCor(\text{criteria}_c, \text{criteria}_{c\prime})$	The conflicting relationships between criteria c and criteria c'
$dCov(\text{criteria}_c, \text{criteria}_{c\prime})$	The distance covariance between criteria c and criteria c'
$dVor(\text{criteria}_c)$	The distance variance of criteria c. $dVor(\text{criteria}_c) = dCov(\text{criteria}_c, \text{criteria}_c)$
$dVor(\text{criteria}_{c\prime})$	The distance variance of criteria $c\prime$ $Vor(\text{criteria}_{c\prime}) = dCov(\text{criteria}_{c\prime}, \text{criteria}_{c\prime})$
e_i	CO_2e emissions at terminal i ($kgCO_2e$)
$f_p(x)$	The objective function value of objective function p
$f_q(x)$	The objective function value of objective function q
$f(x^*)$	The objective function value of new Pareto solution
$\overline{f}_p(x^*)$	Projection of new Pareto solution for objective function p
h_i	Transshipment cost at terminal i (Baht/ton)
I_c	The information content contained in criterion c
$K_q(u_n)$	The objective function in first stage of the two-stage mathematical programs

L	List of rectangles
l_n	Lower bounds of iteration n
o_{ij}^m	CO_2e emissions from terminals i to j by transportation mode m when $i \neq j$ ($kgCO_2e$)
$o\prime_{ij}^m$	CO_2e emissions from terminals i to j by transportation mode m ($kgCO_2e$)
OP_r	The overall performance score for each alternative route
$Q_p(u_n)$	The objective function in the second stage of the two-stage mathematical programs
R_n	Rectangle of iteration n
s_i	Service time at terminal i (hours)
SD_c	The standard deviation of criterion c
t_{ij}^m	Transportation time from terminals i to j by transportation mode m when $i \neq j$ (hours)
$t\prime_{ij}^m$	Transportation time from terminals i to j by transportation mode m (hours)
u_n	Upper bounds of iteration n
u_{np}	The upper bound of objective functions p for iteration n
V_n	The volume measure associated with all rectangles for iteration n
W_c	The objective weight of criterion c
WE_r^*	The positive weighted Euclidean distance
WE_r^-	The negative weighted Euclidean distance
x_{ij}^m	Binary decision variables: 1 = if the goods are transported from terminals i to j by transportation mode m and transshipped at terminal i; 0 = otherwise
x^*	The optimal solution of two-stage formulations $K_q(u_n)$ and $Q_q(u_n)$
y	Corresponding objective vector
$\overline{y}$	Projection of corresponding objective vector
y_p^l	The lower vertex of a rectangle for dimension p
y_p^u	The upper vertex of a rectangle for dimension p
z^*	The optimal objective value of subproblem $K_q(u_n)$.
0	Origin terminal
β^*	Positive ideal solutions
β^-	Negative ideal solutions
ε	ε-constraint value
σ	Large number (σ = 999,999)
γ_N	The Pareto solutions list
$\mathbb{R}^{P-1}$	The search space for $P-1$

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